

## **ELASTIC LOAD BALANCER**

An Elastic Load Balancer in Amazon Web Services is a service that automatically distributes incoming application traffic across multiple targets, such as EC2 instances, to ensure high availability, scalability, and fault tolerance.

### **Types of Elastic Load Balancer in Amazon Web Services**

#### **Application Load Balancer (ALB)**

- Works at **Layer 7 (Application Layer)**
- Used for **web applications**
- **Protocols:** HTTP, HTTPS

#### **Network Load Balancer (NLB)**

- Works at **Layer 4 (Transport Layer)**
- Used for **high performance and low latency**
- **Protocols:** TCP, UDP, TLS

#### **Classic Load Balancer (CLB)**

- Works at **Layer 4 & Layer 7**
- Old/legacy type
- **Protocols:** HTTP, HTTPS, TCP, SSL

#### **Gateway Load Balancer (GWLB)**

- Works at **Layer 3 (Network Layer)**
- Used for **security appliances**
- **Protocols:** IP (uses GENEVE protocol)

## **OSI MODEL WITH PROTOCOLS**

### **Application Layer**

- Used for user interaction (apps)
- **Protocols:** HTTP, HTTPS, FTP, SMTP, DNS

### **Presentation Layer**

- Data formatting, encryption, compression

- **Protocols/Formats:** SSL/TLS, JPEG, PNG, MP4

### Session Layer

- Manages sessions/communication
- **Protocols:** NetBIOS, RPC

### Transport Layer

- End-to-end communication, reliability
- **Protocols:** TCP, UDP

### Network Layer

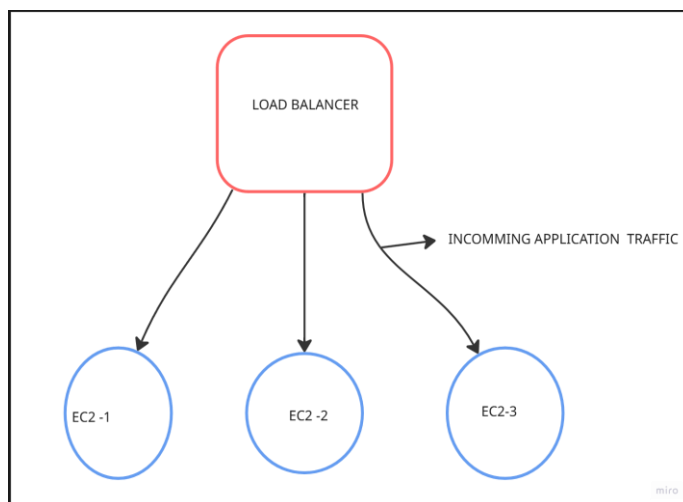
- Routing and IP addressing
- **Protocols:** IP, ICMP, IPSec

### Data Link Layer

- Node-to-node transfer
- **Protocols:** Ethernet, ARP, PPP

### Physical Layer

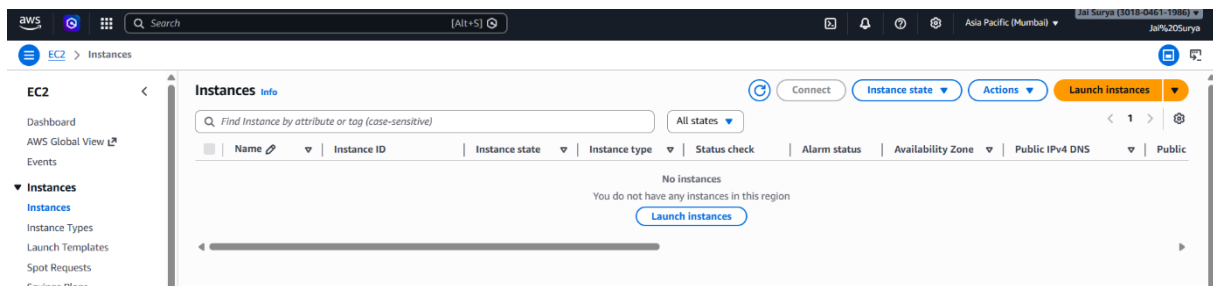
- Physical transmission of data
- **Examples:** Cables, Switch ports, Signals



# HANDS-ON STEPS TO CREATE A CLASSIC LOAD BALANCER

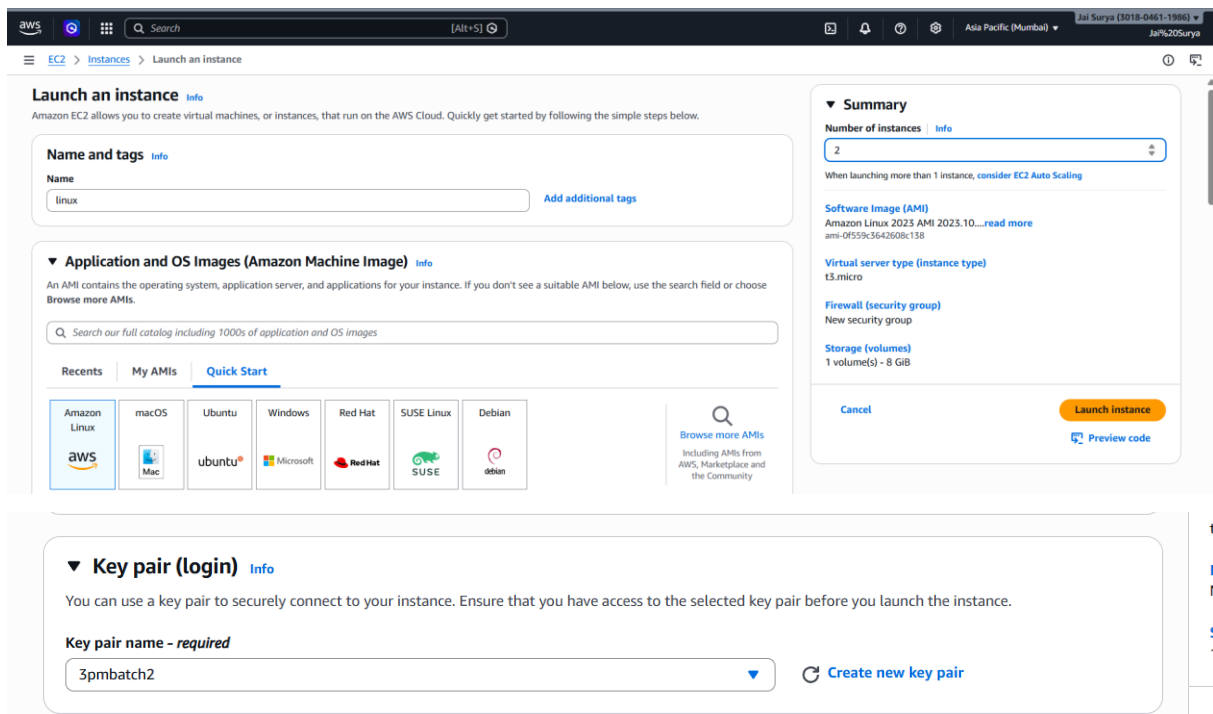
## Screenshot: Launch Instance

- Go to Amazon Web Services Console
- Open EC2 Dashboard
- Click Launch Instance



## Screenshot: Create Instance

- Enter instance name
- Select Amazon Linux AMI
- Choose Key Pair
- Set number of instances to 2



## Screenshot: Configure Network Settings

- Allow SSH (port 22)
- Allow HTTP (port 80)
- Allow HTTPS (port 443)

**▼ Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

SpmBatch2 [Create new key pair](#)

**▼ Network settings** [Info](#) [Edit](#)

**Network** [Info](#)  
vpc-00e925f7d875b3505

**Subnet** [Info](#)  
No preference (Default subnet in any availability zone)

**Auto-assign public IP** [Info](#)  
Enable

**Firewall (security groups)** [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-6' with the following rules:

- ☒ Allow SSH traffic from [Helps you connect to your instance](#) Anywhere (0.0.0.0/0)
- ☒ Allow HTTPS traffic from the internet [To set up an endpoint, for example when creating a web server](#)
- ☒ Allow HTTP traffic from the internet [To set up an endpoint, for example when creating a web server](#)

**▼ Summary**

**Number of instances** [Info](#)  
2  
When launching more than 1 instance, consider EC2 Auto Scaling

**Software Image (AMI)**  
Amazon Linux 2023.10...[read more](#)  
ami-0f553636a7688c138

**Virtual server type (instance type)**  
t3.micro

**Firewall (security group)**  
New security group

**Storage (volumes)**  
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

## Screenshot: Advanced Details

- Go to Advanced Details
- Paste user data script in User Data
- Click Create / Launch Instance

**▼ Advanced details** [Info](#)

**Domain join directory** [Info](#)

**User data - optional** [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash

# Update package repository and install Apache
sudo yum update -y
sudo yum install httpd -y

# Start the Apache and enable it to start on boot
sudo systemctl start httpd
sudo systemctl enable httpd

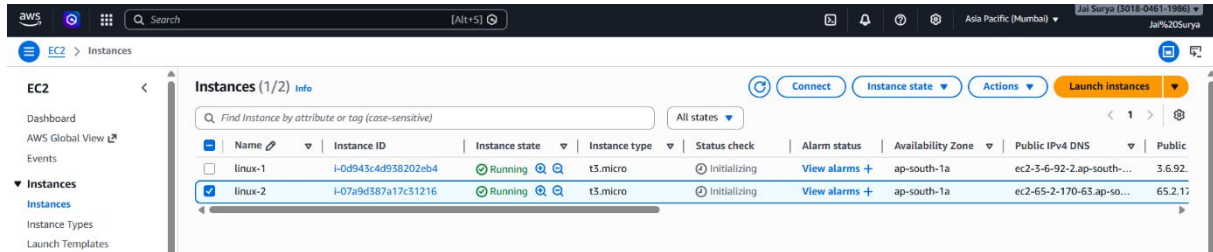
# Create an index.html file with the hostname
echo "<html><head><title>welcome to $(hostname)</title></head></body>
<h1> hello from $(hostname)</h1></body></html>" | sudo tee /var/www/html/index.html

# Restart apache to apply changes
```

☐ User data has already been base64 encoded

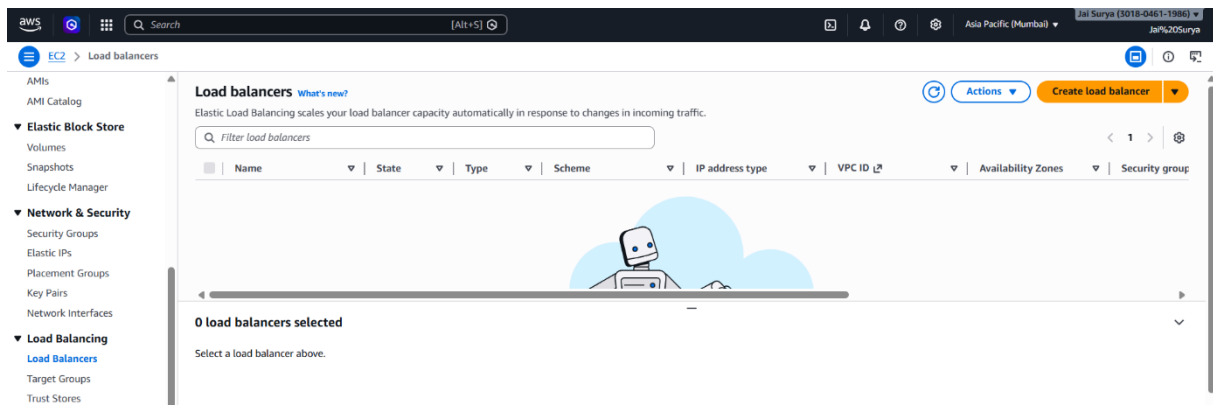
## Screenshot: Instances Created

- Verify two instances are created
- Rename both instances



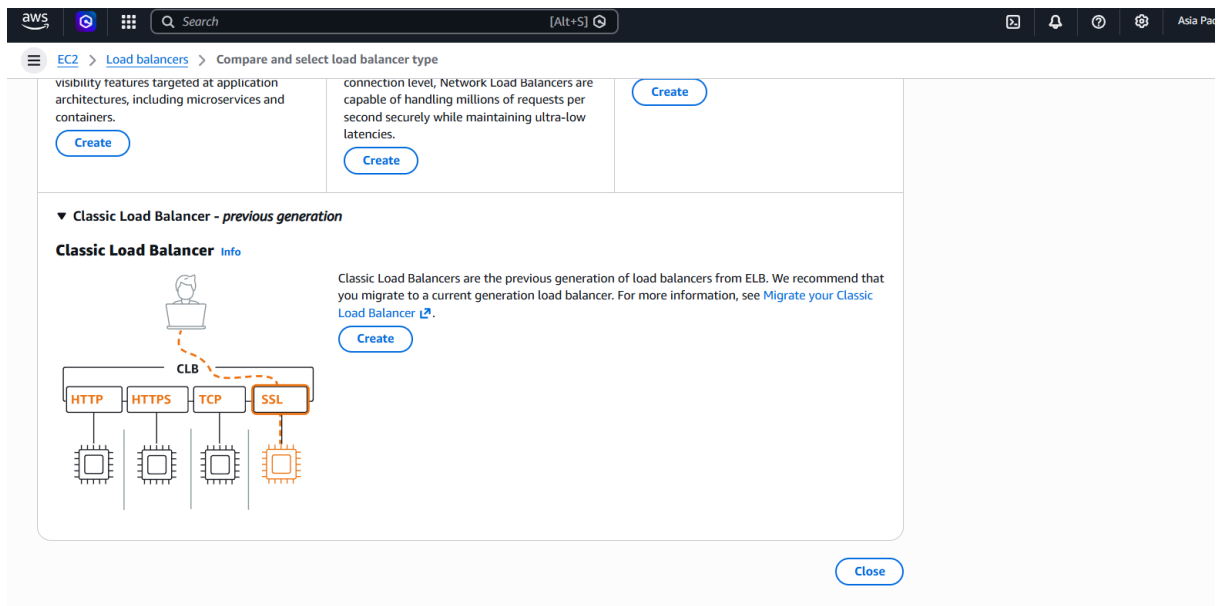
## Screenshot: Go to Load Balancer

- Go to Load Balancers
- Click Create Load Balancer



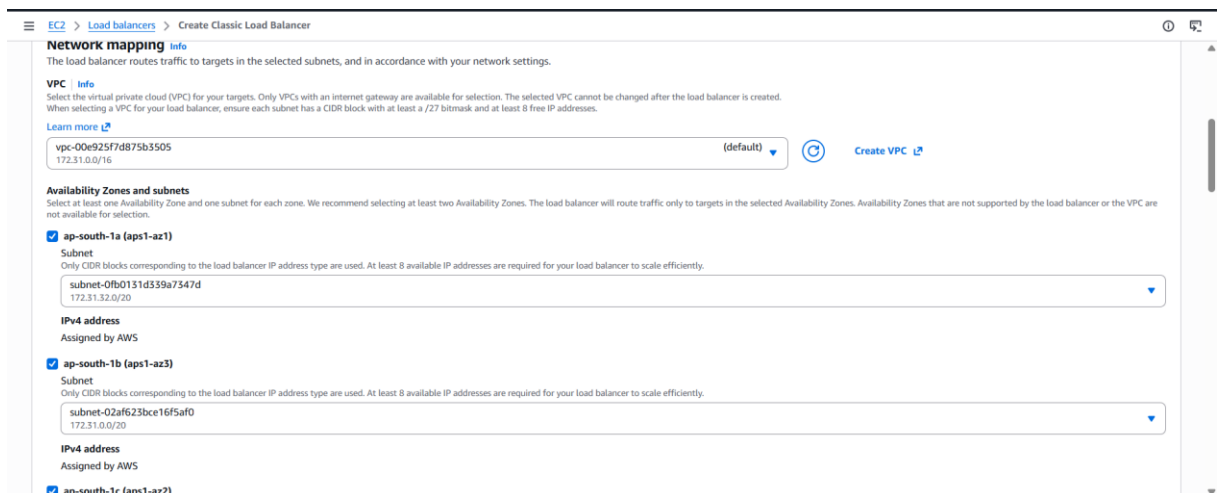
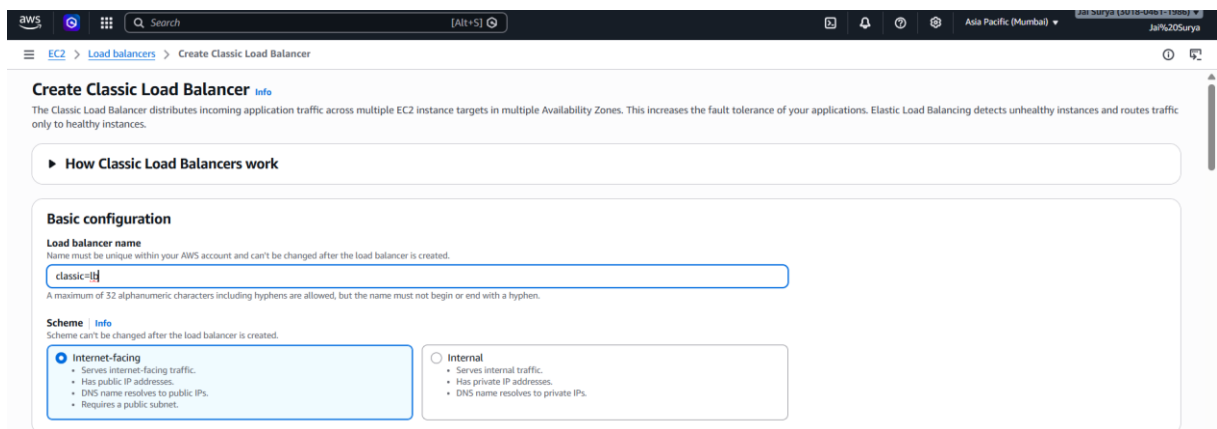
## Screenshot: Select Classic Load Balancer

- Choose Classic Load Balancer
- Click Create



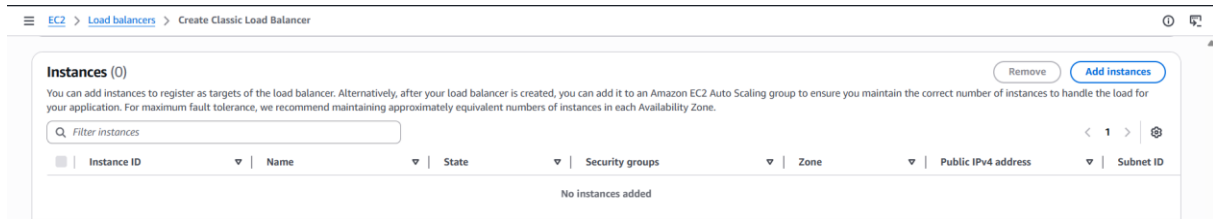
## Screenshot: Configure Load Balancer

- Enter name
- Select Internet-facing
- Select all subnets



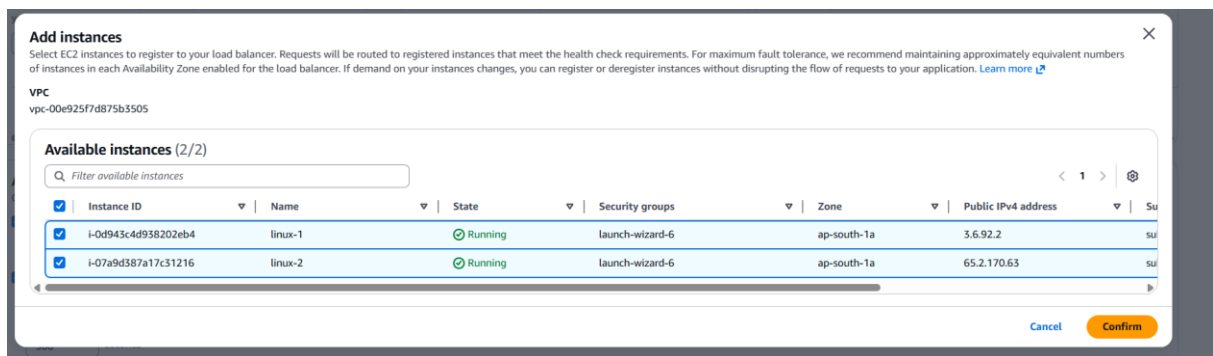
## Screenshot: Add Instances

- Click **Add Instances**



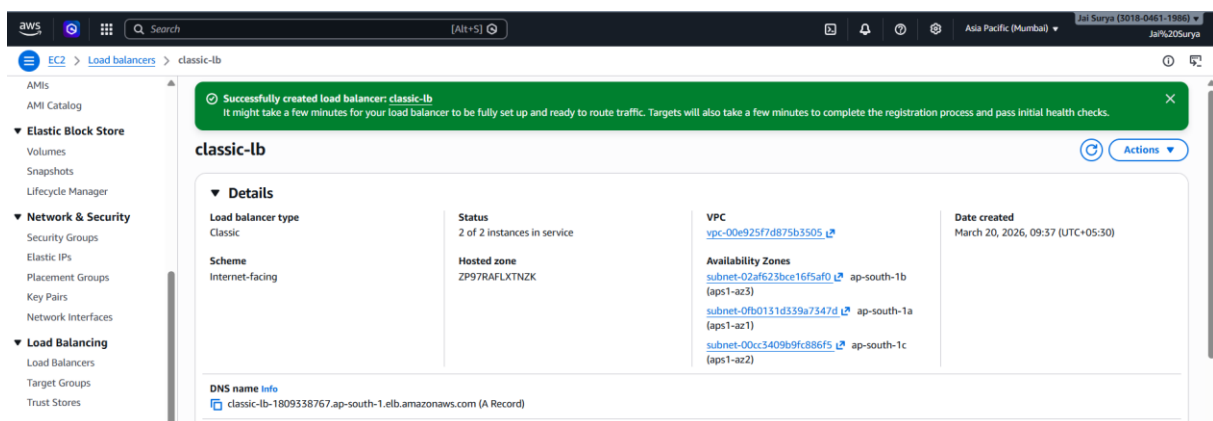
## Screenshot: Select Instances

- Select the two created instances
- Click **Confirm / Add**



## Screenshot: Copy DNS Name

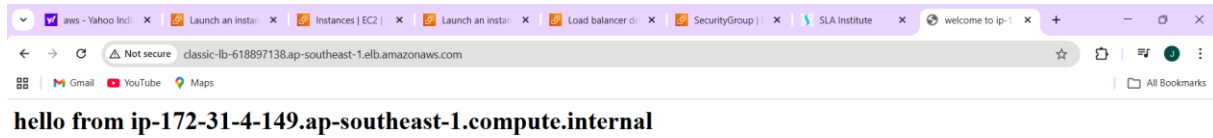
- Go to **Load Balancer details**
- Copy the **DNS name**



## Screenshot: Access Application

- Paste the DNS name in browser

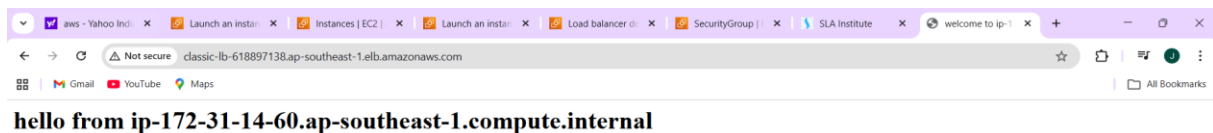
- Press Enter
- Verify the application is accessible



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### Screenshot: Verify Load Balancing

- Refresh the browser page multiple times
- Observe that different instance hostnames are displayed
- Verify traffic is distributed between both instances





## APPLICATION LOAD BALANCER (ALB)

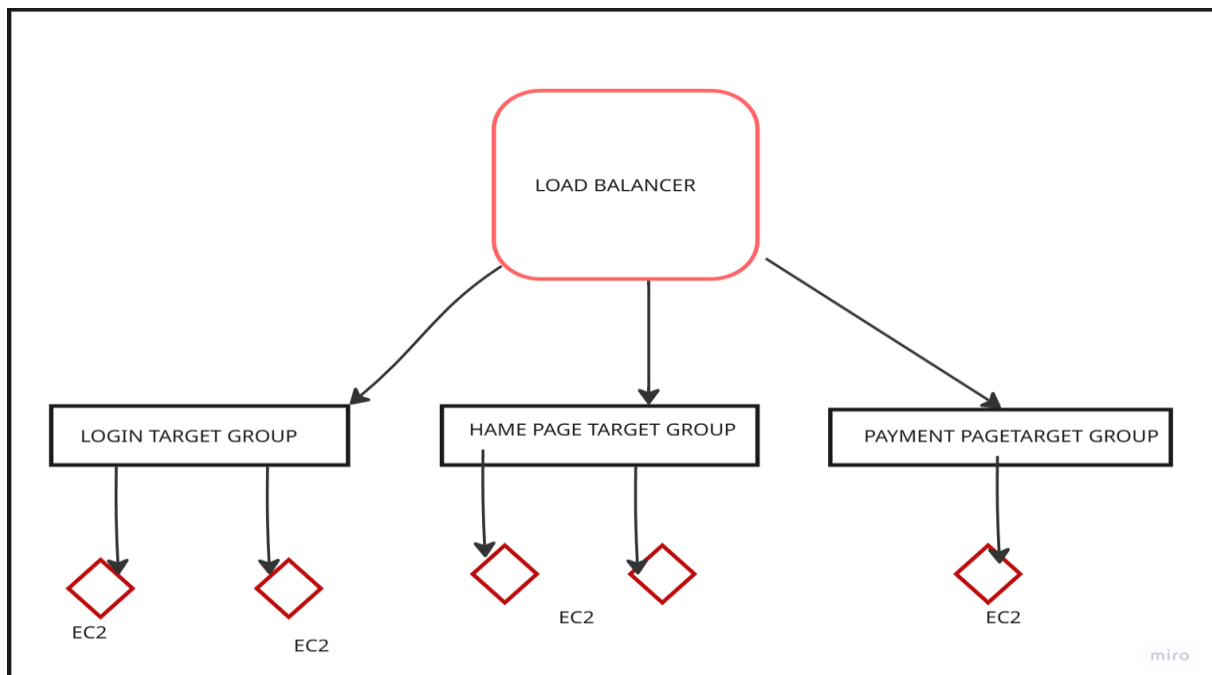
An Application Load Balancer in **Amazon Web Services** is a **Layer 7 load balancer** that distributes incoming HTTP and HTTPS traffic across multiple targets based on request content such as URL paths and hostnames.

### Target Group

A Target Group is a logical group of resources (such as EC2 instances) that receives traffic from a load balancer and distributes it based on defined rules and health checks.

### EC2 Instance

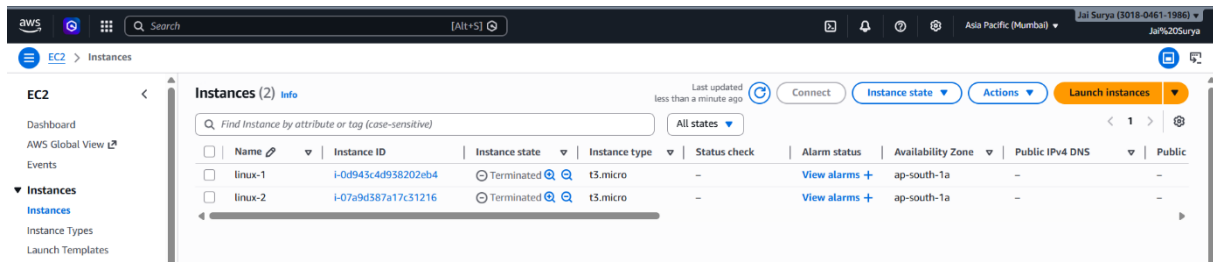
An EC2 instance is a virtual server in AWS used to run applications, which can be registered as a target to receive traffic from a load balancer.



## HANDS-ON STEPS FOR APPLICATION LOAD BALANCER (ALB)

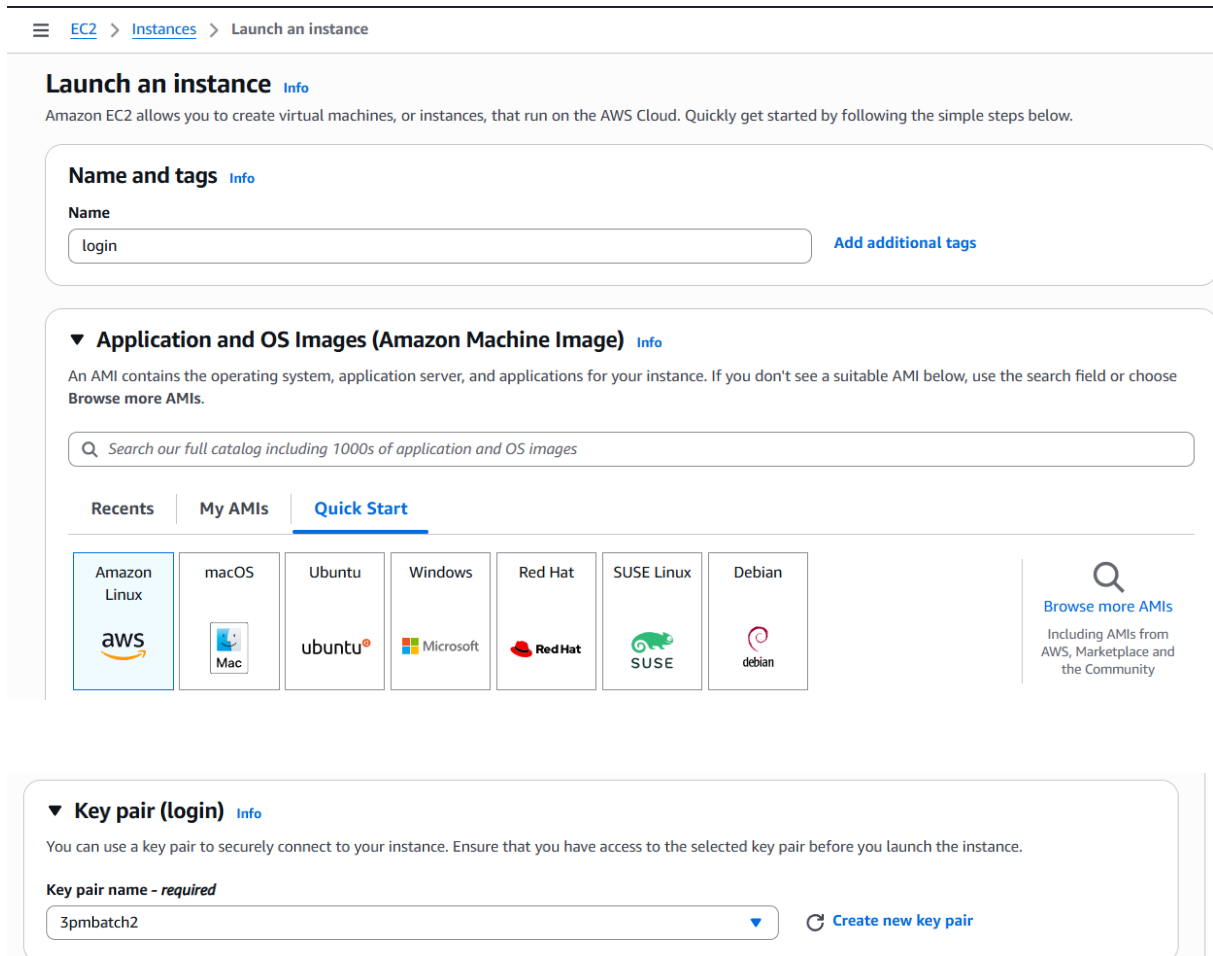
### Screenshot: Launch Instance

- Go to Amazon Web Services Console
- Open EC2 Dashboard
- Click Launch Instance



## Screenshot: Create Instance for Login Page

- Enter instance name
- Select Amazon Linux AMI



## Screenshot: Add User Data (Login Page)

- Go to Advanced Details
- Paste login page script in User Data

User data - optional

Choose file

```
#!/bin/bash

# Update package repository and install Apache
sudo yum update -y
sudo yum install httpd -y

# Start the Apache and enable it to start on boot
sudo systemctl start httpd
sudo systemctl enable httpd

mkdir /var/www/html/login
echo "this is my login page" > /var/www/html/home/index.html

# Restart apache to apply changes
sudo systemctl restart httpd
```

☐ User data has already been base64 encoded

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

ami-0f559c3642608c138

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

## Screenshot: Create Instance for Home Page

- Enter instance name
- Select Amazon Linux AMI
- Paste home page script in User Data

User data - optional

Choose file

```
#!/bin/bash

# Update package repository and install Apache
sudo yum update -y
sudo yum install httpd -y

# Start the Apache and enable it to start on boot
sudo systemctl start httpd
sudo systemctl enable httpd

mkdir /var/www/html/home
echo "this is my home page" > /var/www/html/home/index.html

# Restart apache to apply changes
sudo systemctl restart httpd
```

☐ User data has already been base64 encoded

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

ami-0f559c3642608c138

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

## Screenshot: Instances Created

- Verify two instances are created

EC2

Instances

Find instance by attribute or tag (case-sensitive)

All states

Connect

Instance state

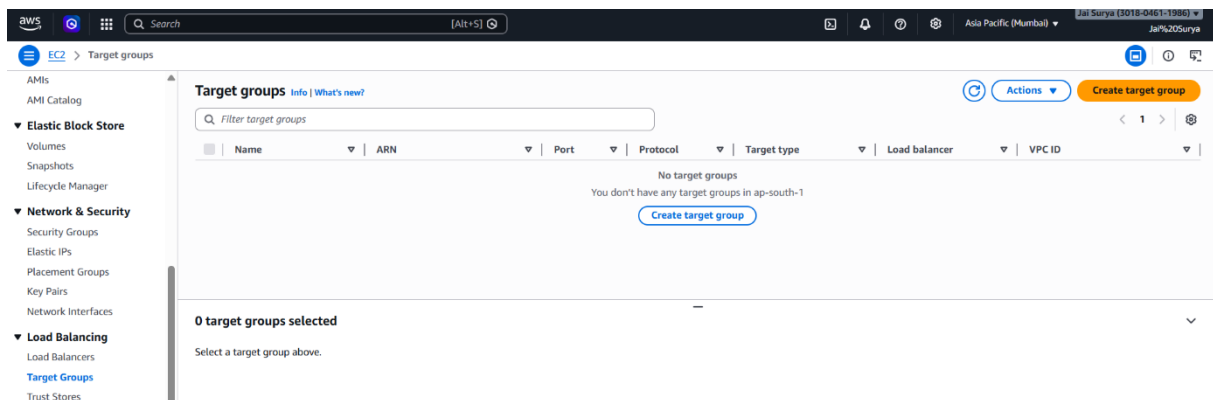
Actions

Launch instances

|                                     | Name    | Instance ID         | Instance state | Instance type | Status check | Alarm status  | Availability Zone | Public IPv4 DNS          | Public  |
|-------------------------------------|---------|---------------------|----------------|---------------|--------------|---------------|-------------------|--------------------------|---------|
| <input checked="" type="checkbox"/> | login   | i-0194c06fb29d254be | Running        | t3.micro      | Initializing | View alarms + | ap-south-1a       | ec2-65-2-170-50.ap-so... | 65.2.17 |
| <input type="checkbox"/>            | linux-1 | i-0d943c4d938202eb4 | Terminated     | t3.micro      | -            | View alarms + | ap-south-1a       | -                        | -       |
| <input type="checkbox"/>            | linux-2 | i-07a9d387a17c31216 | Terminated     | t3.micro      | -            | View alarms + | ap-south-1a       | -                        | -       |
| <input checked="" type="checkbox"/> | home    | i-05170cb0343f3bb27 | Running        | t3.micro      | Initializing | View alarms + | ap-south-1a       | ec2-13-233-44-195.ap-... | 13.233  |

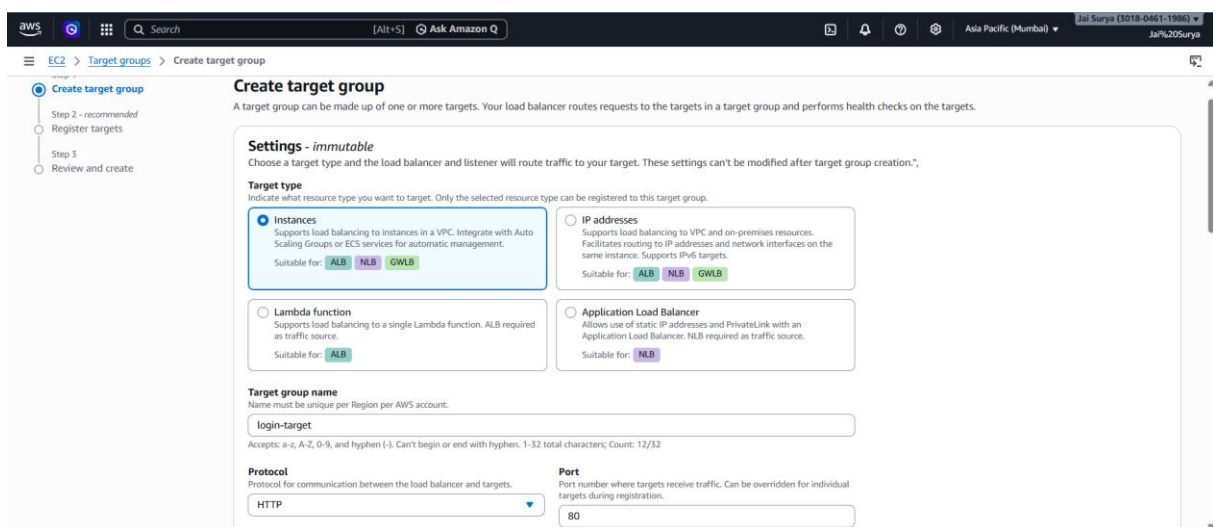
## Screenshot: Create Target Group

- Go to Target Groups
- Click Create target group



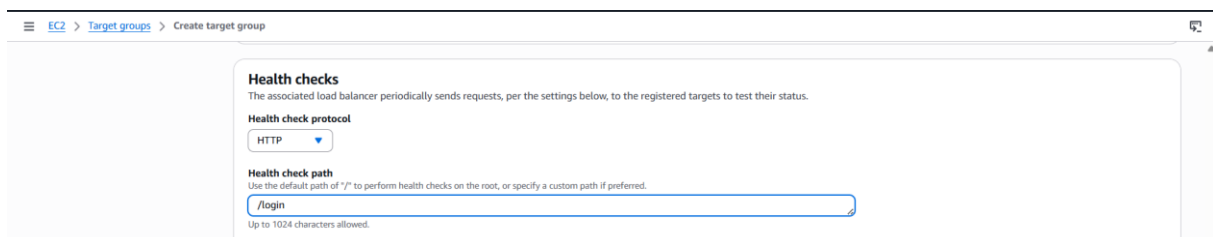
## Screenshot: Configure Target Group

- Select Instances
- Enter name
- Select protocol **HTTP**



## Screenshot: Configure Health Check

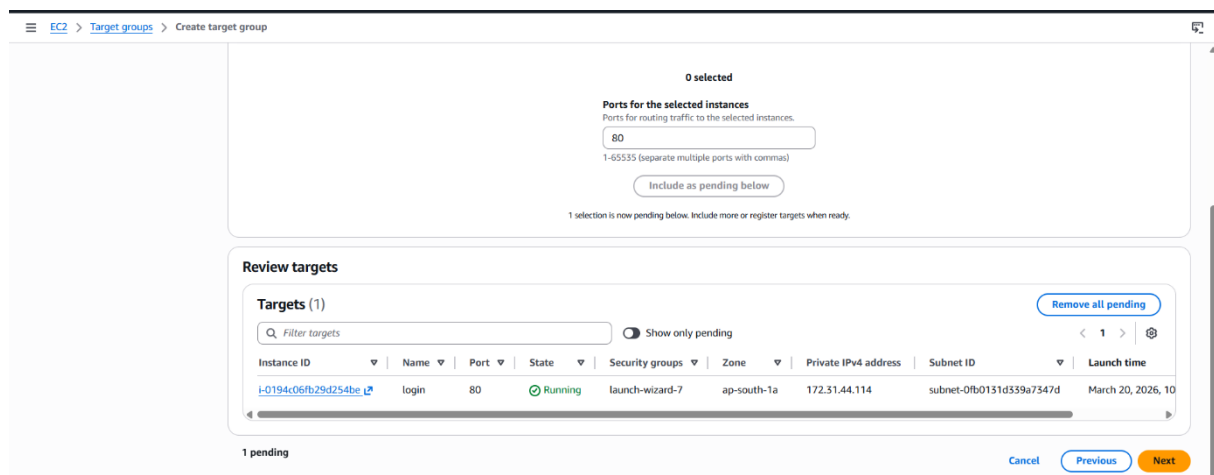
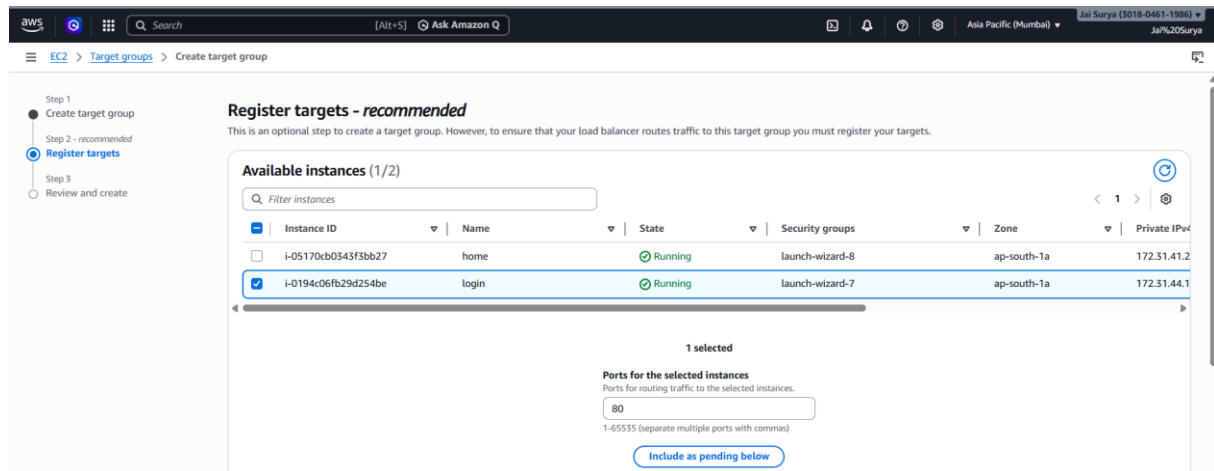
- Set health check path to **/login**



## Screenshot: Register Login Instance

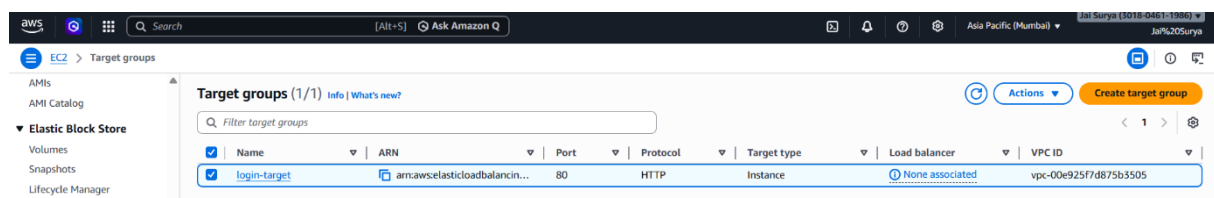
- Select login page instance

- Click Add to registered



## Screenshot: Create Target Group

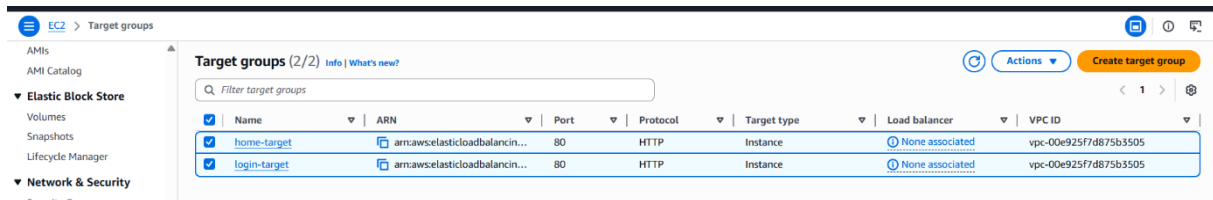
- Click Create target group



## Screenshot: Create Home Target Group , Target Groups Created

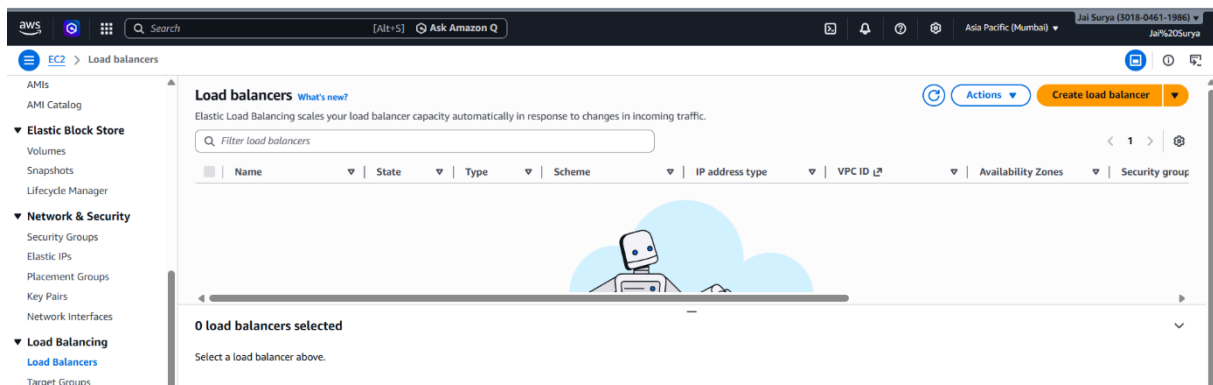
- Repeat the same steps

- Enter name for home target group
- Select protocol **HTTP**
- Set health check path (e.g., / **home**
- Select home page instance
- Click **Create target group**
- Verify both target groups are created



## Screenshot: Go to Load Balancer

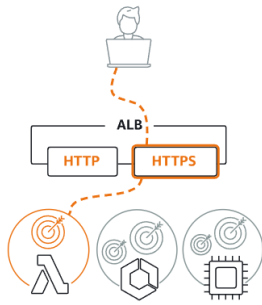
- Go to Load Balancers
- Click **Create Load Balancer**



## Screenshot: Select Application Load Balancer

- Choose **Application Load Balancer**
- Click **Create**

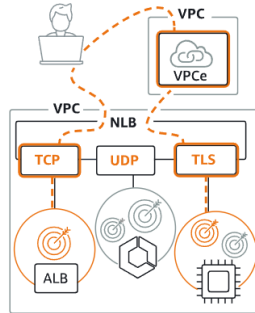
### Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

### Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

### Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

## Screenshot: Configure Load Balancer

- Enter name
- Select Internet-facing
- Select all subnets

### Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

#### ► How Application Load Balancers work

#### Basic configuration

##### Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

application-123

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

##### Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

##### ☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

##### ☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

##### Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

##### ☒ IPv4

Includes only IPv4 addresses.

##### ☐ Dualstack

Includes IPv4 and IPv6 addresses.

EC2 > Load balancers > Create Application Load Balancer

**IP pools** new  
 You can optionally choose to configure an IPAM pool as the preferred source for your load balancer IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).  
☐ Use IPAM pool for public IPv4 addresses  
 The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

**Availability Zones and subnets** Info  
 Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **ap-south-1a (aps1-az1)**  
 Subnet  
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.  
 subnet-0fb0131d339a7347d  
 172.31.32.0/20

☒ **ap-south-1b (aps1-az3)**  
 Subnet  
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.  
 subnet-02af623bce16f5af0  
 172.31.0.0/20

☒ **ap-south-1c (aps1-az2)**  
 Subnet  
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.  
 subnet-00cc3409b9fc886f5  
 172.31.16.0/20

**Security groups** Info  
 A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).  
**Security groups**  
 Select up to 5 security groups  
 default (sg-02d620cc3320d040d)

## Screenshot: Add Target Groups

- In Target Groups section, click **Add target group**
- Add both **login** and **home** target groups

**Target group**

| Target group                                                         | Protocol | Weight | Percent | Action |
|----------------------------------------------------------------------|----------|--------|---------|--------|
| login-target<br>Target type: Instance, IPv4   Target stickiness: Off | HTTP     | 1      | 50%     | Remove |
| home-target<br>Target type: Instance, IPv4   Target stickiness: Off  | HTTP     | 1      | 50%     | Remove |

0-999

**+ Add target group**  
 You can add up to 5 more target groups.

**Target group stickiness** Info  
 Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.  
☐ Turn on target group stickiness

## Screenshot: Load Balancer Created

- Verify that the load balancer is created successfully

aws [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Jai Surya (3018-0461-1986) Ja%20Surya

EC2 > Load balancers

**Load balancers (1/1)** What's new? Actions Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

|                                     | Name           | State           | Type        | Scheme          | IP address type | VPC ID                | Availability Zones   | Security groups      |
|-------------------------------------|----------------|-----------------|-------------|-----------------|-----------------|-----------------------|----------------------|----------------------|
| <input checked="" type="checkbox"/> | application-lb | Provisioning... | application | Internet-facing | IPv4            | vpc-00e925f7d875b3505 | 3 Availability Zones | sg-02d620cc3320d040d |

## Screenshot: Access Login Page

- Copy public IP of login instance
- Paste in browser
- Add **/login** to URL
- Press Enter and verify page



EC2 > Instances > i-0194c06fb29d254be

**EC2**

- Dashboard
- AWS Global View
- Events
- ▼ **Instances**
- Instances

**Instance summary for i-0194c06fb29d254be (login)** Info

Updated less than a minute ago

**Instance ID**  
i-0194c06fb29d254be

**Public IPv4 address**  
65.2.170.30 | [open address](#)

**IPv6 address**  
-

**Instance state**  
Running

Copy public IPv4 address to clipboard



## Screenshot: Access Home Page

- Copy public IP of home instance
- Paste in browser
- Add **/home** to URL
- Press Enter and verify page



## CHANGING PORT NUMBER

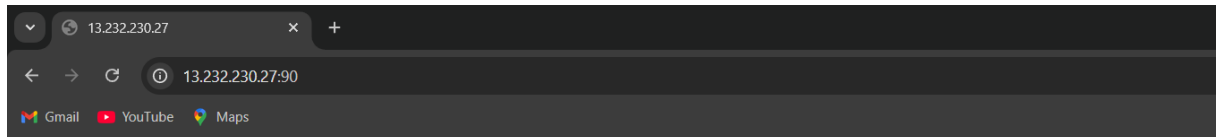
### Open linux terminal execute the webhosting commands

```
Mar 20 10:15:59 ip-172-31-37-76.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 20 10:15:59 ip-172-31-37-76.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 20 10:15:59 ip-172-31-37-76.ap-south-1.compute.internal httpd[25706]: Server configured, listening on: port 80
[root@ip-172-31-37-76 ec2-user]# history
 1 yum update -y
 2 yum update -y
 3 yum install httpd
 4 systemctl status httpd
 5 systemctl start httpd
 6 systemctl status httpd
 7 history
[root@ip-172-31-37-76 ec2-user]#
```

### Running in port number 80



### Change port number 90 shows error



#### This site can't be reached

13.232.230.27 refused to connect.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)

ERR\_CONNECTION\_REFUSED

Reload

Details

### Execute this command

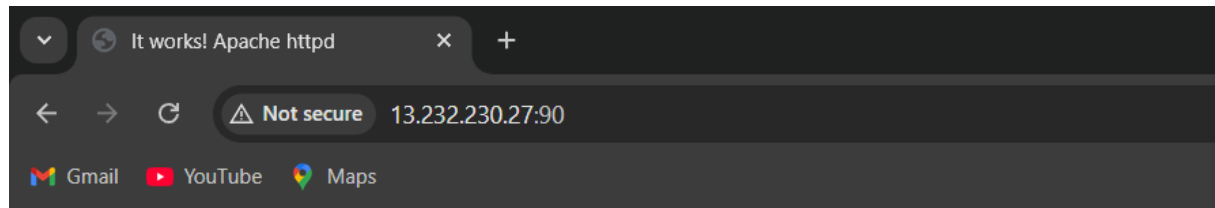
```
[root@ip-172-31-37-76 ec2-user]# nano /etc/httpd/conf/httpd.conf
[root@ip-172-31-37-76 ec2-user]# nano /etc/httpd/conf/httpd.conf
[root@ip-172-31-37-76 ec2-user]# nano /etc/httpd/conf/httpd.conf
[root@ip-172-31-37-76 ec2-user]# systemctl restart httpd
[root@ip-172-31-37-76 ec2-user]#
```

Change port number 90 instead of 80 in listen

```
# available when the service starts. See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:80
Listen 90

#
# Dynamic Shared Object (DSO) Support
#
# To be able to use the functionality of a module which was built as a DSO you
```

Run successfully after changed the port number to 90



It works!